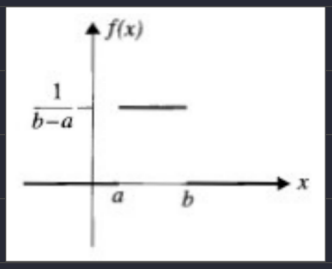
Special Continuous Distributions

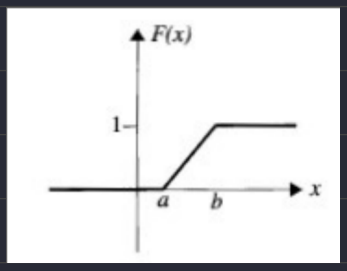
7.1 Uniform Random Variable

Definition of Uniform R.V.

在 (a, b) 區間中的機率都是一樣的，且 uniform = the same

$$f(x) = F'(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases} \qquad F(x) = \begin{cases} 0 & x \le a \\ \frac{x-a}{b-a} & a \le x < b \\ 1 & x \ge b \end{cases}$$





$$E(X) = \int_a^b x \frac{1}{b-a}\,dx = \frac{1}{b-a}\left[\frac{1}{2}x^2\right]_a^b = \frac{1}{2(b-a)}(b^2-a^2) = \frac{a+b}{2}$$

$$E(X^2) = \int_a^b x^2 \frac{1}{b-a}\,dx = \frac{1}{b-a}\left[\frac{1}{3}x^3\right]_a^b = \frac{1}{3}(a^2+ab+b^2)$$

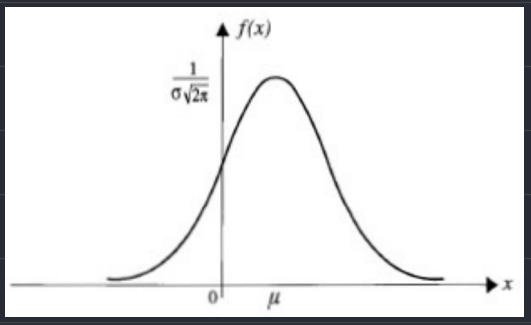
$$\therefore \mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{(b-a)^2}{12}$$

7.2 Gaussian (or Normal) R.V.

Definition

$N(\mu, \sigma^2)$ and the probability density funcion is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



Mean value μ

Variance σ^2

Standard Deviation σ

Standard Normal R.V.

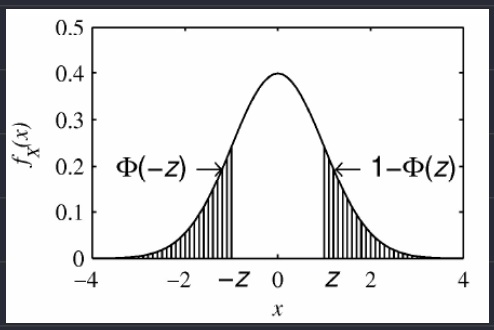
A standard normal random variable Z is a normal random variable with parameters

N (0 , 1) $\mu = 0 \quad \sigma^2 = 1$

p.d.f: $$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

c.d.f: $$P(Z < t) = \Phi(t) = \frac{1}{2\pi} \int_{-\infty}^{t} e^{-\frac{z^2}{2}} dz$$

Note: $\Phi(-z) = 1 - \Phi(z)$



Theorem (Standardized)

For any Gaussian random variable with parameters $N(\mu, \sigma^2)$

the CDF is $F(t) = P(X \le t) = \Phi(\frac{t-\mu}{\sigma})$

and the probability in (a, b] is $P(a < X \le b) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$

Theorem

If X is a Gaussian random variable with parameters $N(\mu, \sigma^2)$

then $E[X] = \mu \quad Var(X) = \sigma^2$ and $Z = \frac{X-\mu}{\sigma}$ is a standard normal R.V

Theorem

If X is a Gaussian random variable with parameters $N(\mu, \sigma^2)$

then $Y = aX + b$, then Y is a Gaussian R.V with parameter $N(a\mu + b, a^2\sigma^2)$

De Moivre-Laplace Theorem

For a Binomial random variable X with parameters B(n, p)

(1) $$\lim_{n\to\infty} P(a < \frac{X - np}{\sqrt{np(1-p)}} < b) = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-\frac{t^2}{2}} dt$$

$\mu = np \quad \sigma = \sqrt{np(1-p)}$ → 標準化後收斂到 $N(0,1)$

(2)

$$P(k_1 \le X \le k_2) \approx \Phi\left(\frac{k_2 + 0.5 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k_1 - 0.5 - np}{\sqrt{np(1-p)}}\right) \quad k_1, k_2 \text{ are integer}$$

→ 為了讓連續的 Normal R.V. 可以再離散的 Binomial R.V. 可以被計算

所以 +0.5 & -0.5 開闢了一條 x 軸空間，避免近似後機率直接 = 0

7.3 Exponential Random Variable

Definition

A continuous random variable X is an exponential random variable with parameter $\lambda > 0$, x > 0 and:

p.d.f. $f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$ arrive rate: λ

c.d.f. $F(t) = \begin{cases} 1 - e^{-\lambda t} & t \geq 0 \\ 0 & t < 0 \end{cases}$ arrive time: t

→ This R.V. is used to estimiate the time between two event

$$E(X) = \frac{1}{\lambda}$$
$$E(X^2) = \frac{1}{\lambda^2}$$
$$Var[X] = \frac{1}{\lambda^2}$$

Ex If the duration of a phone call is an exponential r.v. with mean of 10 minutes. If someone arrives immediately ahead of you at a phone booth, what is the probability that you have to wait

(1) more than 10 minutes ?

(2) between 10 and 20 minutes ?

$E(X) = \frac{1}{\lambda} = 10 \quad \lambda = \frac{1}{10}$

$P(X > 10) = 1 - F(10) = 1 - (1 - e^{-10 \cdot \frac{1}{10}}) = e^{-1}$

$P(10 < X < 20) = F(20) - F(10) = 1 - e^{-20 \cdot \frac{1}{10}} - (1 - e^{-10 \cdot \frac{1}{10}}) = e^{-1} - e^{-2}$

Relation of Exponential and Geometric R.V.

If X is an exponential random variable, then [X], the integer part of X; i.e, the greatest integer less than or equal to X, is a geometric random variable.

→ 作業有證明

Theorem (Memoryless Property)

Expotentional R.V. 會忘記目前的狀態，直接計算條件過後的機率

$P(X \geq t + s \mid X \geq t) = P(X \geq s)$ 已使已經過了 t 年，但是活到 t+s 年的機率不受 t 影響